

YOHANE KUMWENDA

Information Systems and Data Analyst | AI/ML Developer | UNDP Digital Advocate

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PROFESSIONAL PROFILE

I am a final-year BSc Information Systems student at the University of Malawi with a GPA of 3.45 and above. I have practical experience in data analytics gained during my internship at a hospital, where I worked with real clinical and operational data. Beyond my studies, I have built several projects involving machine learning, web applications, and database systems. I am also a trained UNDP Digital Advocate, helping communities across Malawi learn and use digital technology. I enjoy finding practical solutions to real problems through technology and I am ready to contribute meaningfully to any team.

WORK EXPERIENCE

Data Analyst Intern

July 2025 to September 2025

Katete Mission Hospital, Mzuzu, Mzimba District, Malawi

Cleaned, checked, and organised raw clinical and operational records across hospital departments, improving the quality of data available for decision making.

Built Power BI dashboards that allowed management to monitor hospital performance in real time, reducing reliance on manual reports.

Prepared clear analytical reports that helped hospital management make informed decisions on staffing, resources, and operations.

Handled all patient data carefully and in full compliance with GDPR data protection requirements.

Worked closely with the IT team to identify and resolve system problems, keeping operations running without unnecessary interruption.

Studied workflow data to identify areas that were causing delays and recommended practical improvements to management.

VOLUNTEER WORK

Digital Advocate

2026 to Present

UNDP Inclusive Digital Transformation for Malawi (IDT4M) Programme

Worked as a certified UNDP Digital Advocate, delivering digital literacy training to community members across Malawi.

Helped people in communities develop basic and practical technology skills that they can use in their daily lives.

Contributed to national efforts to reduce the digital divide and make technology more accessible to all.

Prepared training materials in local language to make content easier to understand and more relevant to participants.

KEY PROJECTS

NthakaGuide — AI Crop and Fertilizer Advisor

2026

Python, scikit-learn, Flask, PostgreSQL, React

Built an AI-powered web system that recommends the most suitable crops and fertilizer plans for smallholder farmers in Malawi, based on their soil data and local seasonal rainfall.

Trained a Random Forest machine learning model and connected it to a web interface through a Flask API, making the recommendations accessible through any browser.

The system was designed to give rural farmers access to data-driven guidance that can help improve harvests and support food security.

Health Facility Finder System

2024 to 2025

React, Python, PostgreSQL, pgRouting

Developed a web system that helps people in Malawi locate the nearest health facility using their current location and map-based routing.

The system serves both patients searching for nearby clinics and hospital staff who need to manage their service coverage areas.

Voting Management System

2025

React, Python, PostgreSQL

Built a web platform that any organisation can use to run its own elections digitally in a secure and structured way.

The system includes user role management, ballot creation, and live result tracking to support transparent electoral processes.

Dayire Kicks Hub — E-Commerce Platform

2025

React, Python, PostgreSQL

Developed a complete online shop for a Malawian sneaker brand, allowing customers to browse products by size, style, and brand.

Added in-app notifications so users are informed about new arrivals and ongoing promotions.

Examination Attendance Management System

2023 to 2024

University of Malawi

Designed and built a digital system to record and manage student attendance during university examinations, replacing the previous paper-based approach.

Took responsibility for database design and the user interface, which led to fewer administrative errors and saved significant time during exam periods.

Accommodation Management System

2023

Team Project using Agile Methodology

Worked as part of a team to build a web system that manages the allocation of student accommodation at university level.

Contributed across all stages of the project including system analysis, database design, development, and quality assurance testing.

CERTIFICATIONS

Cybersecurity and Its Ten Domains <i>Coursera, 2026</i>	Network Defense <i>Cisco Networking Academy, 2026</i>
Cyber Threat Management <i>Cisco Networking Academy, 2026</i>	Network Support and Security <i>Cisco Networking Academy, 2026</i>
Information Systems Auditing, Controls and Assurance <i>Coursera, 2026</i>	Operating Systems Support <i>Cisco Networking Academy, 2026</i>
Endpoint Security <i>Cisco Networking Academy, 2026</i>	Using Computer and Mobile Devices <i>Cisco Networking Academy, 2026</i>
Introduction to Cybersecurity <i>Cisco Networking Academy, 2026</i>	English for IT 1 <i>Cisco Networking Academy, 2026</i>
Exploring Networking with Cisco Packet Tracer <i>Cisco Networking Academy, 2026</i>	English for IT 2 <i>Cisco Networking Academy, 2026</i>
Foundations of Digital Marketing and E-Commerce <i>Coursera, 2023</i>	JavaScript Essentials 1 <i>Cisco Networking Academy, 2026</i>
Attract and Engage Customers with Digital Marketing <i>Coursera, 2023</i>	Getting Started with Speech in Microsoft <i>Microsoft, 2026</i>
Exploring Internet of Things with Cisco Packet Tracer <i>Cisco Networking Academy, 2026</i>	StartUp Fundamentals <i>Microsoft, 2026</i>
Introduction to Modern AI <i>Cisco Networking Academy, 2026</i>	Trezor Academy Certificate <i>Trezor Academy, 2026</i>
Introduction to Greenhouse Gas Accounting for IT <i>Cisco Networking Academy, 2026</i>	UNDP IDT4M Digital Advocate Certificate <i>UNDP Malawi, 2025</i>

TECHNICAL SKILLS

Data and Analytics	Power BI, Microsoft Excel (Advanced), Data Cleaning and Validation, Statistical Analysis
Programming	Python, JavaScript, React, Flask, HTML and CSS
AI and Machine Learning	scikit-learn, Random Forest, Model Training and Evaluation

Databases	PostgreSQL, SQL, Database Design, Entity-Relationship Modelling, pgRouting
Cybersecurity	Network Defense, Endpoint Security, IS Auditing, Cyber Threat Management
Networking	Cisco Packet Tracer, Internet of Things, Network Essentials, TCP/IP Fundamentals
Design Tools	Figma, Adobe Design Suite, Canva
Platforms	Windows, Linux, Microsoft Office Suite

EDUCATION

BSc Information Systems

Expected: August 2026

University of Malawi (UNIMA) | GPA: 3.45+

Studied core subjects including Database Systems, Software Engineering, Systems Analysis and Design, Human-Computer Interaction, Network Administration, Research Methods, and Information Security.

Independently applied machine learning techniques in the NthakaGuide capstone project using Python and scikit-learn.

Completed 22 professional certifications in areas including cybersecurity, networking, artificial intelligence, digital marketing, and more alongside regular studies.

LANGUAGES

English	Professional working proficiency	Chichewa	Native speaker
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REFERENCES

Eunice Kholowa <i>Lecturer, University of Malawi</i> eunicekholowa@unima.ac.mw	Dr. Martin Thodi <i>Lecturer, University of Malawi</i> martinthodi@unima.ac.mw	Prof. Chipokanjo <i>Lecturer, University of Malawi</i> chipokanjo@unima.ac.mw
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